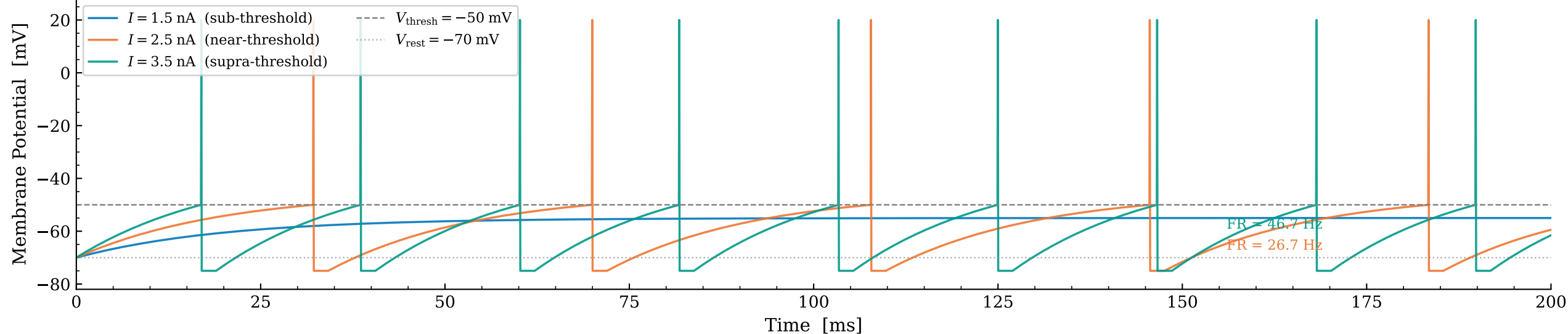
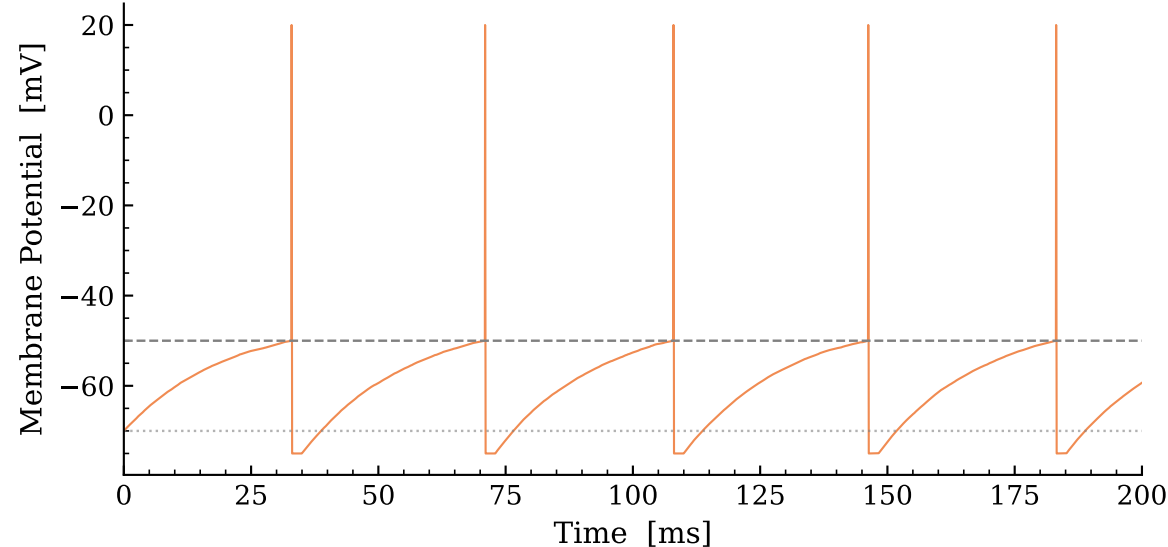


LIF Neuron Simulator: Deterministic · Stochastic (Euler-Maruyama) · Fokker-Planck Benchmark

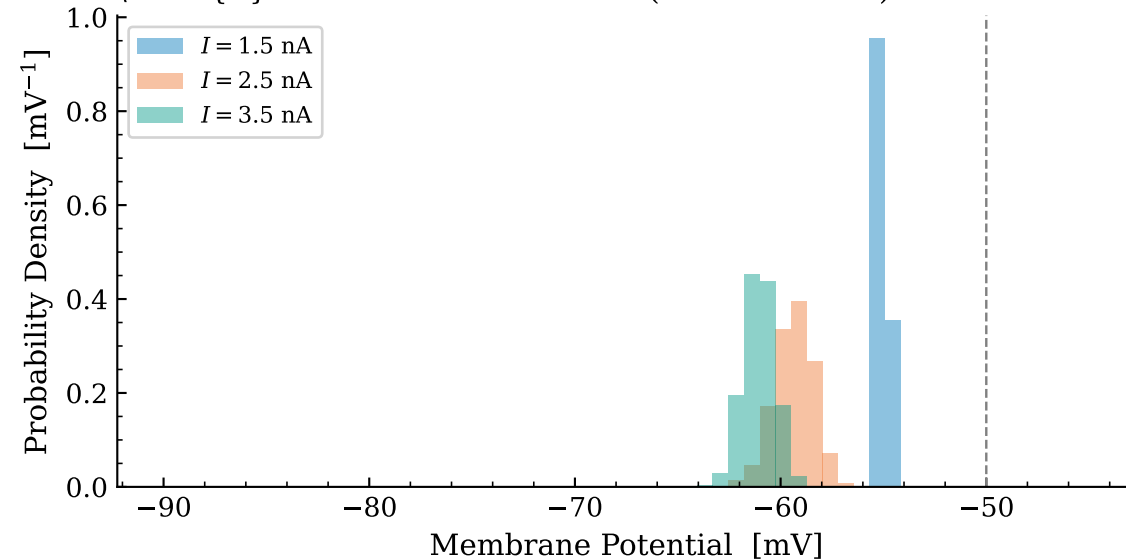
\textbf{A} - Deterministic LIF: voltage traces ($\tau_m = 20$ ms, $R_m = 10$ M Ω)



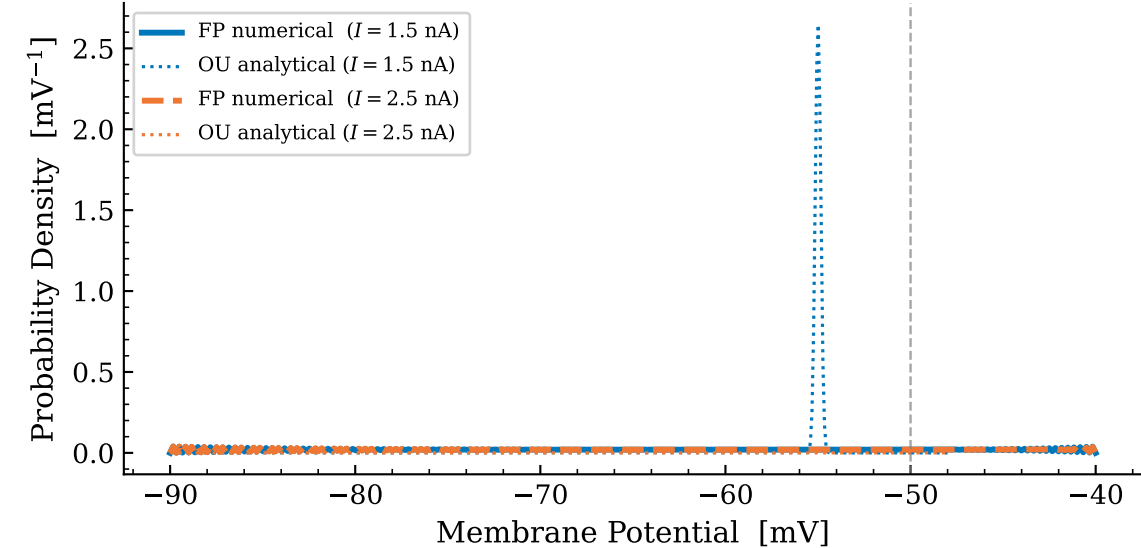
\textbf{B} - Stochastic LIF trace ($I = 2.5$ nA, $\sigma = 1.5$ mV $\cdot$ s $^{-1/2}$)



\textbf{C} - Ensemble distribution (N = 2000 trials)



\textbf{D} - FP stationary PDF: numerical vs. analytical



\textbf{E} - Benchmark: MSE & D_{KL} (analytical OU vs. MC)

